

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) who builds real projects with Claude Code as a daily AI coding assistant, from prompt design through iterative testing and documentation, alongside hands-on financial-services experience at Deutsche Bundesbank. Combines a machine learning and Python/SQL foundation with genuine interest in generative AI and agentic workflows applied to investment and research processes.

Core Competencies

- **Applied GenAI & Agentic Workflows:** Daily hands-on use of Claude Code across a real end-to-end project, prompt design, iterative testing, and documentation; genuine interest in agentic workflows and LLM-based tooling
- **Programming:** Python, SQL, R, Java, C++
- **Financial Services Experience:** Data quality, regulatory reporting, and automated validation at Deutsche Bundesbank (AnaCredit)
- **Machine Learning:** Scikit-Learn, Keras, TensorFlow, Random Forest, predictive modeling, classification
- **Databases & Tools:** SQL, SQLite, MySQL, Git, Jira
- **Analytical Rigor:** Independent research experience (cryptography), structured reporting, German (C1) and English (C1)

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern 04/2026–07/2026
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
 - Built a parameterized Python/SQL validation framework with a YAML-driven configuration layer, replacing hard-coded, case-by-case validation scripts
 - Implemented automated error detection and statistical benchmarking to flag anomalous reporting patterns, with automated ticketing and stakeholder communication
 - Worked within an agile Scrum framework (sprint planning, daily stand-ups, sprint reviews)

Projects

- **Personal Project (GitHub), built with Claude Code**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Designed and built an end-to-end data pipeline for simulated bank transactions (150 accounts), using Claude Code as an AI coding assistant throughout: prompt design, iterative testing, and documentation
 - Built a normalized SQLite database with foreign-key integrity checks and a fully auditable log recording every change
- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity and security parameters of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication
- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Built and validated predictive models (Random Forest, Neural Networks) and K-Means clustering on 25+ years of national health data

Education

- **Tecnológico de Monterrey** **Mexico**
BSc in Data Science *2022–2027*
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, numerical optimization, cryptography & security.
- **Universität Göttingen** **Germany**
Exchange Semester *2025–2026*

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Certifications

- **Python AI: Keras, Scikit-Learn, PyTorch** – LinkedIn Learning (05/2026).
- **Data Science in Python** – University of Michigan, Coursera (04/2026).
- **R Programming** – LinkedIn Learning (05/2026).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.